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Screenless Fracpack with Smart Fracturing System in Cemented Monobore Completion: Offshore Malaysia Case Study

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Abstract

This paper explores the innovative use of screenless fracpack with a smart fracturing system in cemented monobore completion. Innovative frac pack completion is required to developed marginal reserve in fines prone I-reservoirs. This pilot marks the first application in cemented monobore in Asia Pacific. The study delves into design, implementation, execution, production outcomes, and future recommendations, highlighting a cost-effective solution for sand control in challenging environments.

This paper explores the innovative use of screenless fracpack with a smart fracturing system in cemented monobore completion. Innovative frac pack completion is required to developed marginal reserve in fines prone I-reservoirs. This pilot marks the first application in cemented monobore in Asia Pacific. The study delves into design, implementation, execution, production outcomes, and future recommendations, highlighting a cost-effective solution for sand control in challenging environments.

The process began by activating the fracture initiation sub at the bottom of the downhole assembly, allowing the smart dart to be launched and positioned onto the target frac sleeve. This activation occurred at a specific pressure, ensuring the cement sheath outside the tubing was broken with a 3000 psi safety margin above the expected UCS. Once the sub was activated, the dart was launched, opening the frac sleeve for the first zone without difficulty.

The treatment started with a prefrac pumping diagnostic to calibrate the model, including breakdown, step rate tests, and calibration injections. The main fracturing phase followed, using RCP slurry, a gel sweep, and completion brine to displace the treatment. This achieved an effective conductivity of 1,600 md.ft and a frac half-length of 69m. After the screenless fracturing process, the well was left to cure and then carefully unloaded. The production results showed good productivity and effective sand control, with no proppant flowback observed at the surface. This successful implementation demonstrates the potential for cost-effective and efficient sand control in challenging field conditions.

Unlike unconventional applications where overdisplacement of proppants is tolerated, conventional screenless fracpack requires precise volume calculations to ensure optimal underdisplacement, fully propping the near-wellbore with minimal residual proppant. This marks the first use of the smart fracturing system with cemented monobore completion offshore

Introduction

Field B is located in the Malay Basin offshore Peninsular Malaysia. It is situated approximately 200km North East (NE) of Kerteh, Terengganu at water depth of about 70m. The field was discovered in 1975 and since then, five more exploration/appraisal wells were drilled throughout the exploration and appraisal campaign of the field until the year 2010. An infill campaign targeting to develop marginal reserve in fines prone I-reservoirs was embarked in 2023. I-reservoirs consist of thinner sand interbeds within a shale-dominated interval deposited in a fluvial environment with channel facies forming the gas-bearing reservoirs. The primary development target is the I-sand, which includes gas-saturated reservoirs from I-95 to I-122, occurring at depths ranging from 1,200 to 1,450 meters. The formation exhibits an average permeability of 80 mD. Due to the marginal reserves in individual I-sand reservoirs, development is only viable through a multistack reservoir approach, which necessitates a cost-effective solution to ensure economic feasibility. An innovative frac pack completion strategy was piloted as part of this initiative, representing the first deployment of a Smart Fracturing System in a cemented monobore completion within the Asia Pacific region. The pilot aimed to evaluate a cost-effective sand control method tailored for challenging reservoir conditions, with the goal of enabling scalable implementation in future developments.

Overview of Smart Fracturing System

The Smart Fracturing System is designed for a 4-1/2" cemented monobore well with five stages (I-95, I-100, I-105, I-120, and I-122 sand). Each stage utilizes a single point Smart Fracturing System with the fracture initiation sub positioned below the deepest frac sleeve. This system means that the fracturing treatment is performed at one frac sleeve at a time. The fracture initiation sub provides hydraulic communication with the formation during the deployment of the first intelligent dart and prevents hydraulic lock situations.

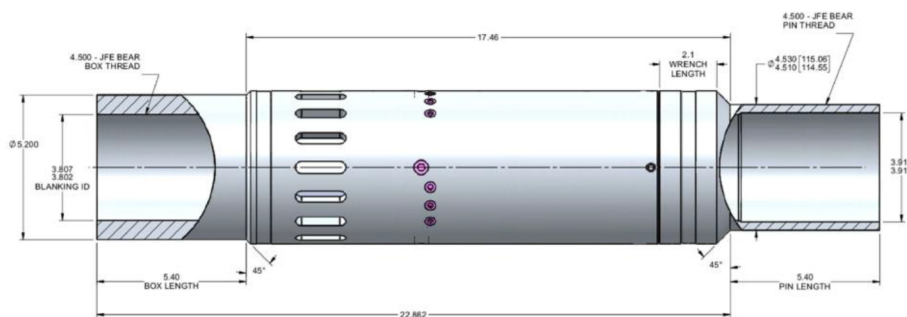


Figure 2—Single Point Frac Sleeve

The differential pressure needed to open the sleeves is adjustable by selecting specific shear screws installed into the sleeves before deployment. For this application, an opening pressure of approximately 2160 psi was chosen, considering the dynamic pressure during the cementing operation. Each single-point sleeve has a fixed area open to flow (AOF) of 9.62 in², achieved through machined slots within the outer sleeve housing. Erosion-resistant materials are fitted at the nozzles to enhance efficiency during the pumping treatment.

The intelligent dart is pumped from the surface at a maximum rate of 4 bbl/min, reducing to 1 bbl/min approximately 10 bbls before seating into its pre-programmed stage. During this process, the dart uses electromagnetic technology to identify and count the frac sleeves as it passes through them, similar to a Casing Collar Locator (CCL) log. The dart uses a countdown system to determine when to expand to the activation OD. When the dart reaches its activation location or counts the second-to-last frac sleeve, the actuator assembly triggers, driving the seal element and slips onto the slip cone. In the activated state, the larger OD dart (3.700") locates and seats in the targeted frac sleeve (3.505"), allowing applied pressure to

open the sleeve. The intelligent dart in this application is made of regular dissolvable material, which loses isolation within 24 hours when exposed to completion brine.

In principle, one frac sleeve is allocated per stage. However, given the pilot nature of this application, contingency measures were implemented by installing backup sleeves to enhance operational reliability. Although there are five stages representing five target zones in the completion string, a total of eight frac sleeves were installed. I-122 and I-105 sand, identified as the primary contributors to production, were equipped with an additional frac sleeve positioned directly above the primary sleeve. This configuration acts as a safeguard against potential stimulation challenges, such as sleeve malfunction or premature screenout. A minimum spacing of 12 meters between the primary and backup frac sleeves is required to ensure proper dart activation prior to seating in the primary sleeve. This distance also ensures the backup sleeve can contain the frac treatment similar to the primary sleeve. In the event of an early screenout, rather than performing a well cleanout using coiled tubing (CT), a more efficient approach is to wait for the dart to either dissolve or lose its isolation capability. Once isolation is lost, a new dart can be deployed to initiate fracturing at the secondary sleeve location, allowing the operation to proceed without intervention. For the top-most sleeve, a dummy sleeve is installed above it to start counting down the number of frac sleeves, as there is no previous sleeve for the intelligent dart to activate itself and seat on the I-95 frac sleeve. Alternatively, a normal solid dart (no electronic) with 3.700" OD can be used to open the top-most sleeve, but this requires the frac cross on the rig floor to be broken off as the OD of the normal dart cannot pass through the frac cross ID. Since the top-most sleeve lacks a preceding sleeve for dart activation, a dummy sleeve is installed above it. This allows the intelligent dart to initiate its countdown sequence and accurately seat on the I-95 frac sleeve. An alternative approach involves using a standard solid dart with a 3.700" OD to open the top-most sleeve. However, this method requires removing the frac cross at the rig floor, as the dart's outer diameter exceeds the internal diameter of the frac cross.

In a cemented monobore well, the frac sleeve is assembled with the completion tubing and cemented in place, providing no adjustability in the sleeve position before the pumping treatment. In contrast, a gravel pack completion using screens allows greater flexibility in selecting frac locations, as reservoir access is achieved through a perforation system. Optimal sleeve placement in cemented monobore wells will be guided by real-time Logging While Drilling (LWD) data, ensuring that fracturing analysis reflects actual reservoir conditions for precise and effective stimulation. As the Smart Frac System employs uniform frac sleeves and intelligent darts, sleeve placement must be optimized based on critical reservoir parameters such as gas saturation, minimum horizontal stress, gas-water contact, and shale barriers to ensure effective fracture containment and improve production efficiency.

In unconventional reservoirs, proppant overflush is a widely adopted practice during multistage fracturing in horizontal wells. This approach addresses challenges posed by extended fracture closure times, often lasting several weeks—combined with horizontal well geometry and the use of low-viscosity fluids, which promote proppant settling near the frac port. Overflushing helps redistribute proppant deeper into the fracture, preventing accumulation at the port and ensuring effective stimulation.



This plot can be used to determine the flow rate required to achieve a desired annular velocity for solids transport. The annular velocity can be used to determine bottom-up time and depends on flow rate and annular cross-sectional area.

- *** To edit/erase text, double click on text to bring up edit box
- *** To move text box, drag with mouse

Annular Velocity - ft/min

Flow Rate - bbl/min

2.00 in , 3.96 in

Flow Rate (bbl/min)	Annular Velocity (ft/min)
0	0
1	~80
2	~160
3	~240
4	~320

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Planning and designing screenless fracpack with a smart fracturing system

Given the characteristic of I-sand Particle Size Distribution, (d50: 39–56 microns), poor sorting (d40/d90: 12–18), and high fines content (40–60%), Enlarged Wellbore/Resin Consolidation or at minimum, Cased Hole Gravel Pack is the recommended sand control method as per Company Holistic Sand Management Guidelines. In the beginning, the well was design as 9-5/8" Cased Hole Gravel Pack for frac pack operation. However, as part of well design optimization, the well design was later optimized to a 4-1/2" cemented monobore design with frac pack.

Candidate Selection

Candidate screening was conducted for wells targeted for completion in the I-sand as part of this pilot application. Well-A9 was selected based on several criteria: no requirement for a resin job due to low permeability values (less than 100mD), higher unconfined compressive strength (UCS) in the deeper I-sand interval (ranging between 1800–2000 psi), targeted sand away from GWC and no anticipated liquid loading issues in the 4-1/2" tubing considering the operational specifications of the Smart Fracturing System.

Several considerations need to be considered when designing the completion string for the Smart Fracturing System in a cemented monobore setup, in order to activate the toe initiation sub and open the frac sleeve. The considerations are summarized below:

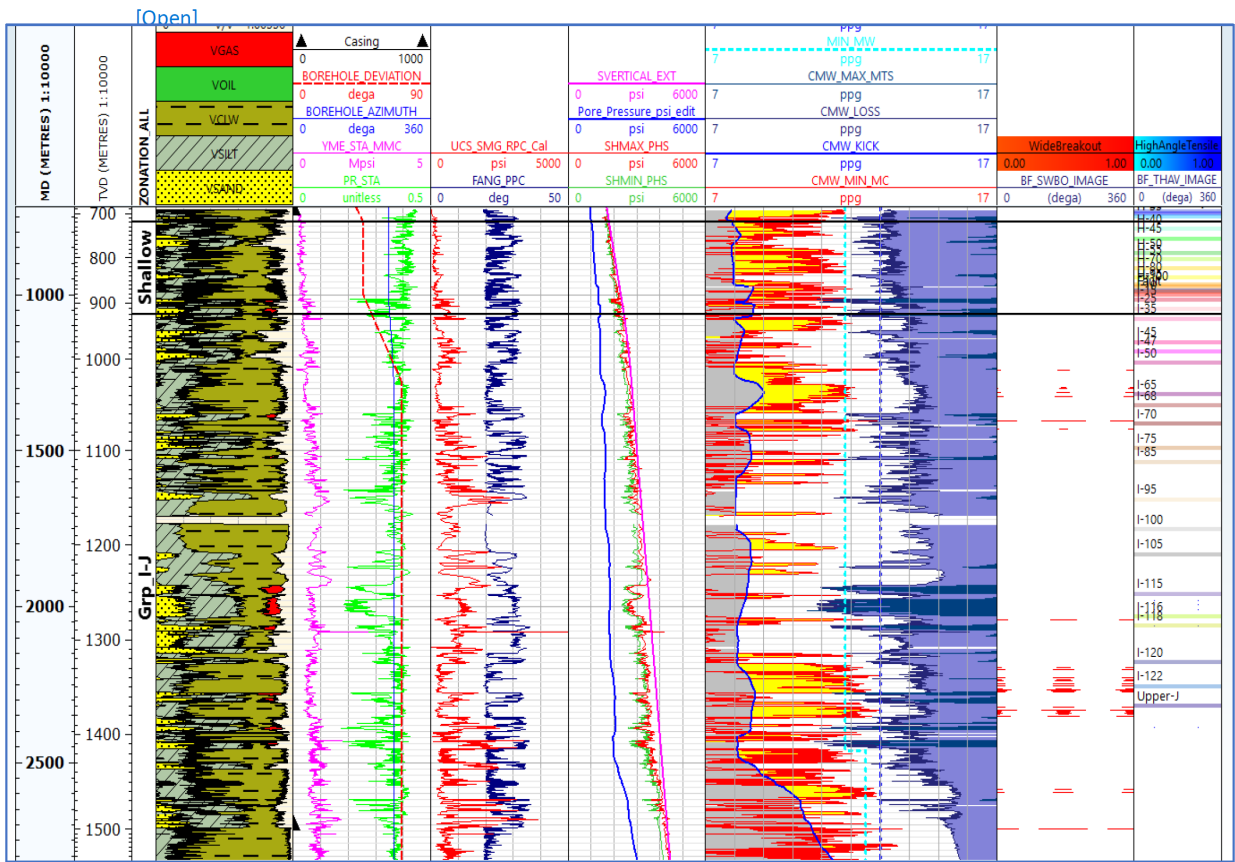


Table 1—Completion Design Consideration

Equipment	Considerations
Christmas Tree	Tree saver must be used to prevent the cavities and valves inside Christmas Tree to be damaged. Use of tree saver will reduce the ID of the Christmas Tree.
Wellhead	Able to sustain high pressure with a minimum of 7,500 psi during fracture initiation sub activation. Isolate the tubing hanger and frac the formation via landing joint.
Subsurface Safety Valve	Cement-through type and able to withstand erosion from frac job.
Permanent Downhole Gauge	Cement-through type and able to withstand erosion from frac job.
Metal Expendable Packer (MEP)	The MEP setting pressure must be low as possible as the fracture initiation sub activation pressure must be above it.
Frac Sleeves	- Has the smallest ID in the well which is 3.505". Slickline and e-line equipment must be smaller than 3.505" to pass through. - Cement evaluation log, tagging top of proppant and setting of bridge plug need to have equipment OD less than 3.505"
Intelligent dart	- Dissolution test needs to be done with the required completion brine to check if the dart can hold pressure during frac treatment until the next stage. - The material of the dart will determine the suitable mill bit for CT during proppant cleanout run.
Fracture Initiation Sub	- Once activated, it will be open permanently and needs to be isolated. - Highest setting pressure in the completion string. - Best to set across sacrificial zone to prevent unwanted hydrocarbon coming into the well.
Cement Wiper Plug	Able to pass through the minimum ID of frac sleeve with maximum 2000 psi to ensure the frac sleeve is not open accidentally during cementing.
Landing Joint & Frac Cross	- Able to launch the intelligent dart with 3.45" OD. - Have a common connection or x-over to slickline, e-line, CT and frac cross. - Need to be broken off multiple time due to rigging up / down slickline, e-line, CT and frac cross in each stage. - Double barrier for well control situation.

Proppant Sizing and Studies

Proppant selection was based on the Saucier method, which recommends sizing at 5–6 times the formation sand's median grain size (d₅₀). Given the I-sand's d₅₀ of 39–56 microns, poor sorting (d₄₀/d₉₀: 12–18), and high fines content (40–60%), a 40/70 mesh resin-coated proppant (210–420 microns) was selected. This ensures effective sand control, minimizes fines migration, and maintains fracture conductivity. Accordingly, further lab testing was conducted to determine the critical velocity for resin-coated proppant (RCP) flowback in this screenless frac pack application.

Laboratory Test for Proppant Flowback. A 40/70 RCP pack is placed between sandpaper and polymer-coated metal plates to simulate the closure stress on the RCP pack. A mixture of heated water and glycerol is pumped through the flowback cell at increasing flow rates, and the differential pressure is measured. Free proppant is collected on a fine mesh metal sieve. From this laboratory test, the critical velocity for RCP flowback, necessary for mathematical modeling and simulations, is determined. Various levels of curing strength of the RCP are prepared, with the "least curing strength" being 65°C for 4 hours and the "strongest curing strength" being 85°C for 24 hours. The RCP pack with the least curing strength is unstable with a low critical velocity, while the RCP pack with the strongest curing strength shows a high critical velocity exceeding 63 cm/s and a high pressure drop at failure. These critical velocities are incorporated into a flowback simulator and modeled with specific geomechanical properties of the reservoir. The results indicate that a longer RCP curing time of over 48 hours, combined with a drawdown duration of at least 6 hours and a maximum drawdown of 300 psi, can prevent proppant flowback and ensure the near-wellbore

area remains propped. For CT cleanout performed in less time than the curing time, it is preferred to conduct it under overbalance conditions. If performed under underbalance conditions, the maximum limit is 75 psi.



Figure 6—Lab test setup for proppant flowback

Frac pack design in cemented monobore

The frac pack design utilized in this cemented monobore application is similar with conventional frac pack operation. The process is outlined below:

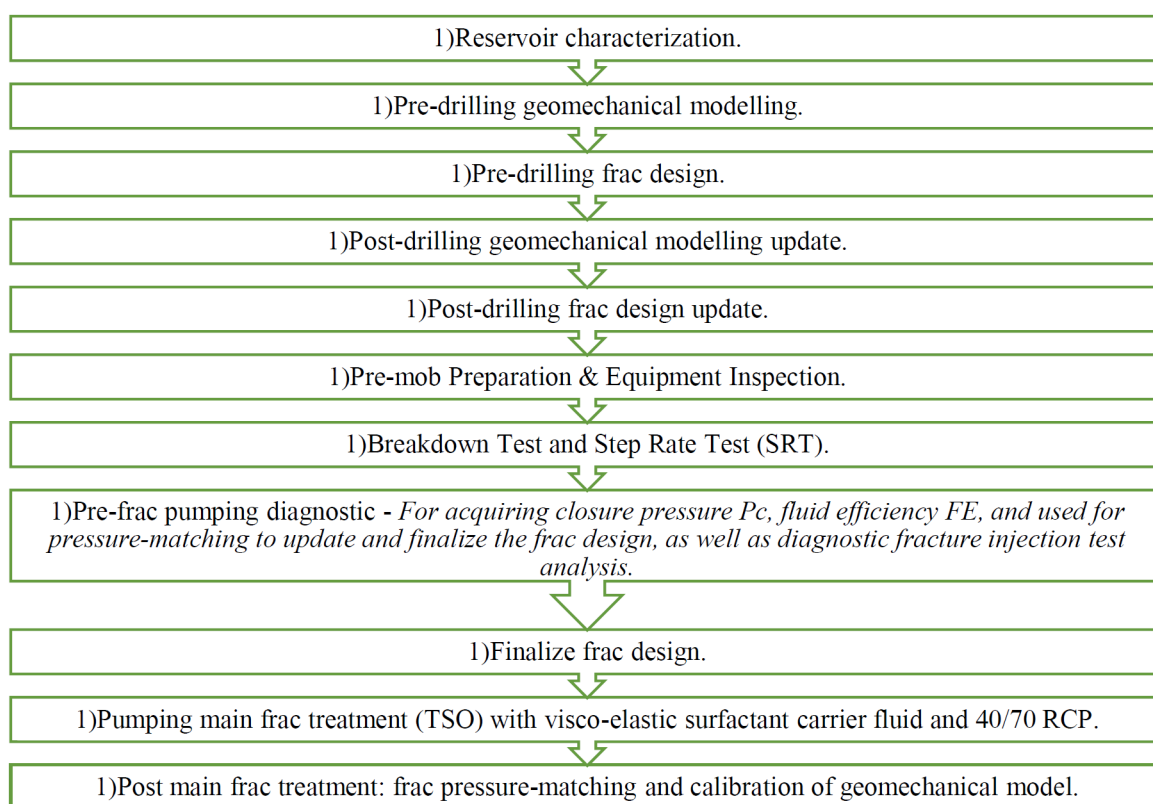


Figure 7—Frac pack design and execution operation

Smart Fracturing System Execution in Cemented Monobore Completion

The well completion started with execution of cemented monobore completion, followed by deployment of smart fracturing system. The process can be summarized as per flowchart below:

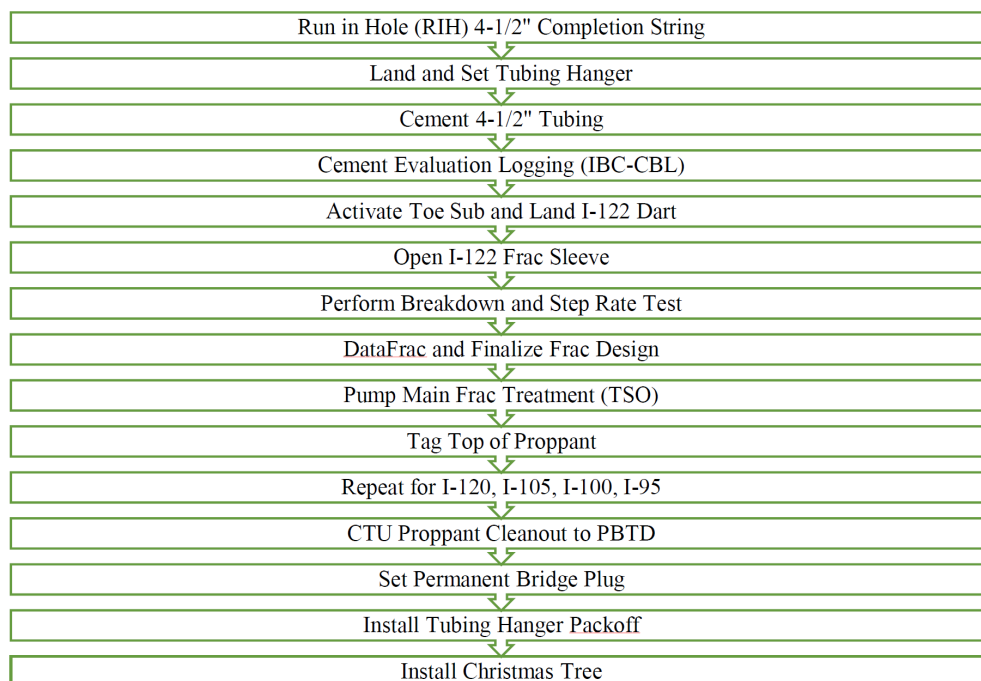


Figure 8—Smart Fracturing System Execution in Cemented Monobore Completion

The fracture treatment design was governed by three critical parameters. To maximize fracture conductivity, a tip screen-out (TSO) strategy was employed. Proppant concentration was capped at 7 lbm/gal to reduce the likelihood of premature screenout, while the pump rate was limited to 20 bbl/min due to equipment capacity and pressure constraints.

Operationally, the sequence began with the activation of the fracture initiation sub—the lowermost component of the downhole assembly—which established a flow path for deploying the dart to the I-122 zone. Following activation, the dart was launched from surface to engage the I-122 frac sleeve, confirmed by the pressure signature shown in Figure 9.

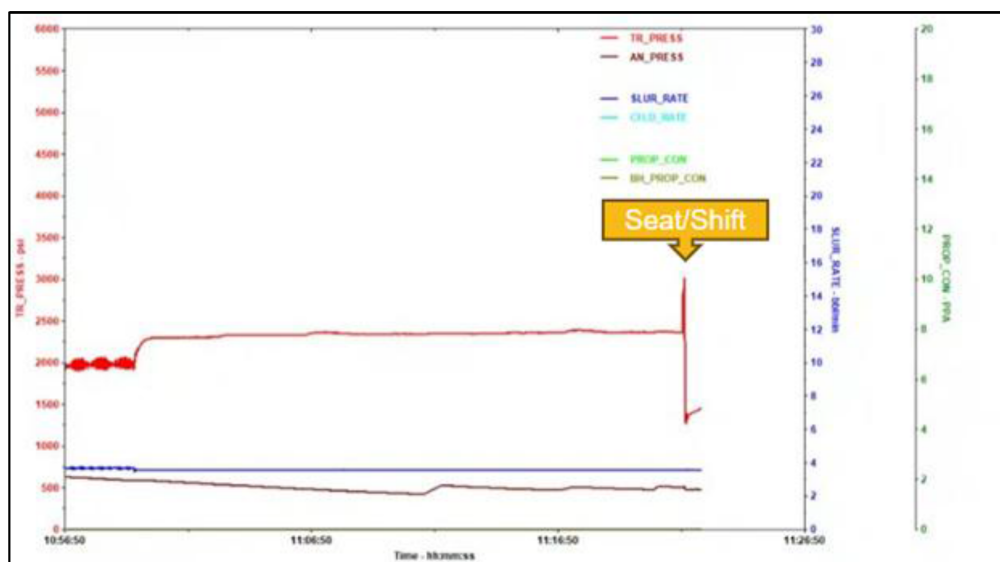


Figure 9—I-122 Dart Landing Signature

I-122 treatment started by performing a pre-frac pumping diagnostic, to calibrate the model involving breakdown, Step Rate Test and Calibration Injection. During Calibration Injection Test, the residual volume of the slurry after the frac pump is stopped was also checked due to the 2 bbls under flush requirement. Stopping the pump later than pumping schedule can severely impact the top of proppant height.

Table 2—I-122 Pre-Frac Summary

I-122 Pre-Frac Summary	P Extension (psi)	Q Extension (bpm)	P Closure – decline (psi)	Efficiency
Breakdown			2951	3.8%
Step Rate Test	3696	6.45	2702	4.2%
Mini Frac			2746	4.4%

Table 3—I-122 Main Frac Summary

I-122 Main Frac Summary		
Total Proppant Pumped	31173	lbs
Frac Sleeve Shifting Pressure	2890	psi
Frac Sleeve Shifting Volume	114	bbls

Upon completion of I-122 treatment, a slickline run was done to tag top of proppant. The slickline tool string encountered an obstruction in the build-up section, with the running weight indicating the possible presence of residual carrier fluid or RCP on the low side of the interval. Further complicating the operation, the tool string could not be pumped down due to insufficient injectivity, as the carrier fluid in well I-122 had not yet degraded. As a result, the decision was made to retrieve the slickline assembly and proceed with a coiled tubing intervention to clean the dart landing area. This was necessary to isolate the I-122 zone and enable treatment of the I-120 zone.

The operation was resumed for the next interval, I-120. The main frac was successfully completed, with 21,393 lbs of proppant pumped, followed by a 15-barrel gel sweep and completion brine to displace the treatment fluids. Post-fracturing, a slickline run was conducted, during which the tool string tagged the dart at the I-105 frac sleeve—significantly shallower than the intended I-120 target. Multiple LIB (Lead Impression Block) attempts confirmed that the dart had mis-landed at the I-105 sleeve, indicating an unanticipated operational complication. This confirmed that the previous treatment, intended for the I-120 sand, was inadvertently applied to the shallower I-105 sand. As the pumping volumes had been calculated based on the I-120 port depth, the fracture was overflushed, resulting in proppant placement further away from the wellbore and into the I-105 sand.

Coiled tubing (CT) was deployed to mill out the dart seated at the I-105 frac sleeve and subsequently run a shifting tool to close the I-122 sleeve. Due to insufficient injectivity preventing dart deployment via conventional means, a final attempt was made to convey the dart to the I-120 frac sleeve using CT. However, RCP remained in the build-up section of the well, resulting in premature dart lodging at the I-105 secondary frac sleeves.



Figure 10—I-120 Dart LIB Impression

Following the successful milling of the dart, the decision was made to transition from the Smart Fracturing System to the plug and perforation method for the remaining four zones, enabling continued progress in the operation. While high well deviation posed a known challenge, potentially leading to sand accumulation along the low side of the wellbore; the team proactively addressed this by pumping 1,358 lbs of proppant as a sand plug via CT. The CT subsequently tagged the top of the sand plug slightly higher than anticipated, overlapping the I-120 frac sleeve. Multiple bullheading flushes using brine and gel were effectively employed to reposition the sand toward the target zone. Ultimately, a physical tag confirmed successful placement of all sand plugs into the I-122 formation, with no residual sand left in the wellbore. Based on this positive outcome, the well was completed in the I-122 zone, and operations were advanced to the next well. The remaining four zones will be revisited in the future phase using an optimized remedial sand control strategy.

The pilot implementation in the I-122 single zone was concurrently evaluated through well unloading and a sustained four-month production period, delivering an average gas rate of 5 MMscf/d without any surface sand production or proppant flow back. Following this, pressure build-up tests were conducted to analyze reservoir performance and quantify the skin factor. Based on the performance, it can be deduced that the pilot is able to meet the objective of delivering I-122 production while controlling the sand production.

Lesson Learnt

The execution of this pilot provided several key insights that are critical for optimizing future operations. A detailed investigation identified four potential root causes or process error mechanisms that could contribute to such events, such as programming or launch order errors, excessive pressure, Electro-Magnetic Interference (EMI) and dart clearance.

Programming and Launch Order Errors

Incorrect programming of the dart for the intended stage and misplacement of programmed darts in a multi-dart launcher can cause this error to happen. Although a launcher was not used in this pilot, strict adherence to programming protocols remains critical; whereby programming documentation was reviewed and confirmed correct dart was correctly programmed for the I-120 primary sleeve. In addition, witnesses and video recordings confirmed that the dart was powered on with the correct count representing the number of stages.

Excessive Pressure

The second root cause identified is excessive pressure that may cause unintended opening of the frac sleeve. Pressure log analysis during the deployment of the I-122 and I-120 darts revealed distinct operational behaviors. The I-122 dart demonstrated a stable pressure profile, with a single pressure spike occurring precisely at the point of landing on the designated frac sleeve. In contrast, the I-120 dart exhibited multiple unstable pressure spikes from the beginning of its run, with the highest spike reaching 6,000 psi. Although this pressure, when combined with hydrostatic pressure, remained below the dart's maximum operating limit of 13,000 psi, the irregular pressure pattern suggests a potential mechanical interference. It is hypothesized that proppant accumulation ahead of the dart, followed by its dislodgement near the sleeve's inner diameter, may have inadvertently triggered the opening of the I-105 sleeve.

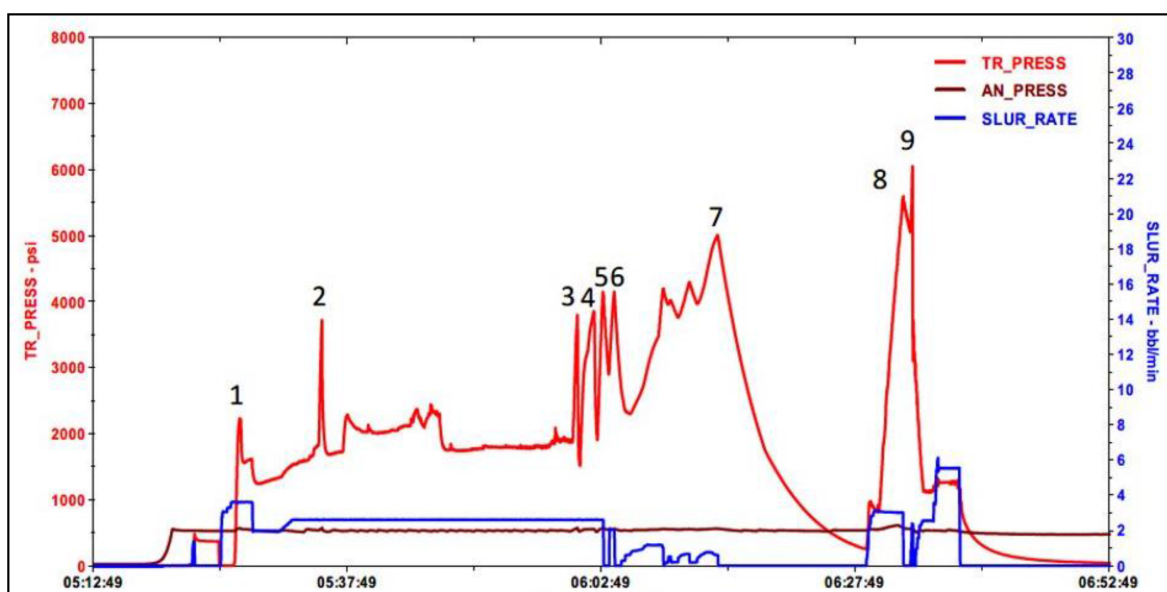


Figure 11—I-120 Dart Landing Signature

Electro-Magnetic Interference (EMI)

External sources of Electro-Magnetic Interference (EMI) with sufficient intensity can cause false or ghost counts after dart activation, potentially compromising sleeve targeting accuracy. An investigation into EMI during and after the operation revealed that the electromagnetic field generated by a 5A current from Permanent Downhole Gauge (PDG) was minimal and not strong enough to influence the dart's ghost count behavior.

Dart Clearance

The clearance between the dart's outer diameter in its inactive state (3.450") and the sleeve's inner diameter (3.505") is minimal—approximately 0.055". This tight tolerance is critical for reliable seating and activation but poses a risk of the dart becoming stuck if residual proppant remains after CT cleanout. Additionally, the well's deviation likely contributed to incomplete proppant removal, as it affected suspension behavior during the CT operation.

Recommendations

Based on the challenges observed during the pilot phase of the Smart Fracturing System, the following recommendations are proposed to enhance operational reliability and efficiency:

- Employ cross-linked gel as the carrier fluid to improve proppant transport and reduce settling.

- Verify pumped volume against dart depth to identify potential hold-ups.
- Perform slickline verification to confirm dart depth before pumping job.
- Adopt a continuous pumping strategy for frac treatment, paired with a specialized launcher for seamless dart deployment.
- Ensure system operability in offline mode to minimize rig time consumption.
- Integrate a noise detector to verify proper dart seating on the frac sleeve.
- It is also recommended for future pilot well candidate selection be selected based on lower angle and fewer zones to be completed.

Conclusion

The pilot implementation of the Smart Fracturing System in a cemented monobore completion offshore Malaysia has demonstrated a viable and cost-effective solution for sand control in marginal, fines-prone reservoirs. This first-of-its-kind application in the Asia Pacific region successfully delivered sustained gas production from the I-122 zone, achieving an average rate of 5 MMscf/d with no observed proppant flowback. The integration of intelligent dart technology, precise sleeve placement guided by LWD data, and tailored proppant design contributed to the operation's success. Operational challenges experienced during this pilot stage provide valuable insights for future optimization. Recommendations such as the use of cross-linked gel, continuous pumping strategies, and enhanced verification protocols are expected to improve system reliability and efficiency. Overall, the pilot validates the potential of screenless fracpack with smart fracturing system in cemented monobore completion to unlock reserves in future developments.

Nomenclature

AOF	= Absolute Open Flow
CCL	= Casing Collar Locator
CT	= Coiled Tubing
CHGP	= Cased Hole Gravel Pack
LIB	= Lead Impression Block
LWD	= Logging While Drilling
PBU	= Pressure Build Up
PDG	= Permanent Downhole Gauge
RCP	= Resin Coated Proppant
TSO	= Tip Screen Out

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